
Oracle APEX Developer Complete Learning Path

From Beginner to Certified APEX Professional

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1. Introduction to Oracle APEX

Oracle APEX (Application Express) is Oracle's enterprise low-code application development platform that runs directly on Oracle Database. It enables developers to build scalable, secure web and mobile applications up to 20x faster with minimal coding. APEX is free with every Oracle Database — no additional licensing required. The current version is APEX 24.2 (GA January 2025, latest patch 24.2.13).

Key Features

- **App Builder** – Visual drag-and-drop application development environment for building pages, regions, and components.
- **SQL Workshop** – Browser-based SQL IDE for creating and managing database objects, loading data, and running queries.
- **RESTful Services** – Create and consume REST APIs declaratively using Oracle REST Data Services (ORDS).
- **Team Development** – Built-in project tracking, bug management, and milestone tracking for collaborative development.
- **AI-Assisted Development (GenDev)** – AI Assistant, RAG sources, Generate Text with AI, and natural language schema creation.
- **Workflows** – Native workflow engine for business process management with modular, reusable workflows.
- **Progressive Web Apps (PWA)** – Build installable, mobile-ready apps with push notifications.
- **Universal Theme & Template Components** – Responsive, accessible UI framework with Theme Roller customization.
- **JSON Data Sources** – Work directly with JSON data from Oracle Database 23ai Duality Views.
- **Vector Search** – Semantic similarity search using Oracle AI Vector Search for RAG-powered applications.

Deployment Options

- **Oracle Cloud** – APEX Service, Autonomous Database, or OCI Compute.
- **On-Premises** – Any Oracle Database (11g and above).
- **apex.oracle.com** – Free workspace for learning and prototyping.

■ **Tip:** APEX is **FREE** with every Oracle Database — no additional licensing required. You can start immediately with a free workspace at **apex.oracle.com**. No credit card, no installation, no setup.

2. Prerequisites

Before starting your Oracle APEX journey, ensure you have the following foundational knowledge:

- **SQL** – Solid understanding of SQL queries, joins, subqueries, DML, and DDL statements.
- **PL/SQL** – Basic knowledge of procedures, functions, packages, and triggers.
- **HTML & CSS** – Basic understanding of web markup and styling for UI customization.
- **JavaScript (optional)** – Helpful for dynamic actions and custom client-side interactivity.
- **Oracle Database Concepts** – Tables, views, sequences, indexes, constraints, and schemas.
- **Web Development Basics** – Understanding of HTTP, REST APIs, and JSON data format.
- **Oracle Cloud Account (optional)** – Free tier available for APEX Service on OCI.

■ **Tip:** *SQL and PL/SQL are the backbone of Oracle APEX. If you're not confident in these areas, invest time in strengthening your skills before diving into APEX development. BEENUM LEARNING offers beginner-friendly courses on Udemy to help you get started.*

3. Phase 1: Foundation (Weeks 1–3)

■ ■ PHASE 1: FOUNDATION • Weeks 1–3

Learning Objectives

- Sign up for a free workspace at **apex.oracle.com** or provision APEX Service on OCI.
- Navigate the APEX development environment: **App Builder, SQL Workshop, Gallery, Workspace Administration**.
- Understand APEX architecture: **Workspace, Schema, Application, Page, Region, Item**.
- Create your first database application from scratch.
- Build applications from **spreadsheets, CSV files, and existing tables**.
- Work with **SQL Workshop**: Object Browser, SQL Commands, SQL Scripts, Quick SQL, Data Workshop.
- Create and manage pages: **Reports, Forms, Charts, Calendars, Maps**.
- Understand **Page Designer**: rendering tree, properties pane, layout pane, and gallery.
- Preview, run, and share applications.

Hands-On Activities

- Create a workspace and build your first "Hello World" application.
- Build an Employee Directory app from an existing database table.
- Import a spreadsheet and auto-generate an interactive application.
- Practice with SQL Workshop: create tables, load data, and run queries.

Resources

- [BEENUM LEARNING – Udemy](#)
- [BEENUM LEARNING – YouTube](#)
- docs.oracle.com/en/database/oracle/apex/24.2/

4. Phase 2: Intermediate (Weeks 4–7)

■ ■ PHASE 2: INTERMEDIATE • Weeks 4–7

Learning Objectives

- Build **Interactive Reports** with customization: search, filter, highlight, pivot, group by, compute.
- Build **Interactive Grids** for inline editing and advanced data manipulation.
- Create **Classic Reports** and **Cards** regions for flexible data display.
- Develop **Forms**: single-row, multi-row, master-detail, and tabular forms.
- Implement **Validations, Computations, and Processes** (page processing pipeline).
- Work with **Dynamic Actions** for client-side interactivity (show/hide, refresh, set value).
- Create and consume **RESTful Web Services** (REST Data Sources).
- Use **Shared Components**: LOVs, Navigation, Templates, Breadcrumbs, and Lists.
- Implement **Application Items** and **Application Processes** for global logic.
- Build **Charts** (bar, line, pie, Gantt, etc.) and interactive **Dashboards**.
- Customize the **Universal Theme** with Theme Roller and custom CSS.

Hands-On Activities

- Build a Task Management app with Interactive Reports and Forms.
- Create a Dashboard with multiple chart types and summary regions.
- Implement Dynamic Actions for show/hide fields, refresh regions, and set values.
- Consume an external REST API and display data in a report.

Resources

- [BEENUM LEARNING – Udemy](#)
- [BEENUM LEARNING – YouTube](#)
- [docs.oracle.com/en/database/oracle/apex/24.2/htmdb/](https://docs.oracle.com/en/database/oracle/apex/24.2/htmldb/)

5. Phase 3: Advanced (Weeks 8–11)

■ PHASE 3: ADVANCED • Weeks 8–11

Learning Objectives

- Implement **Security**: Authentication Schemes (APEX accounts, Social Sign-In, Custom, LDAP, SAML, OCI IAM) and Authorization Schemes (role-based access).
- Build **Progressive Web Apps (PWA)** with push notifications and offline capabilities.
- Implement **Workflows** for business process management: structured, modular, and invoke workflow activities.
- Work with **Faceted Search** and **Smart Filters** for advanced data discovery.
- Master **Plug-ins**: use and create custom plug-ins (region, item, process, dynamic action).
- **JavaScript and jQuery** integration for advanced client-side behavior and custom interactions.
- Custom **CSS and Template Components** for pixel-perfect UI design.
- Implement **Email and PDF** generation using APEX_MAIL, APEX_UTIL, and printing APIs.
- Application **deployment**: export/import, Supporting Objects, CI/CD with SQLcl.
- **Multi-environment management**: DEV → UAT → PROD lifecycle and best practices.
- Build **Approval Pages** and multi-step approval processes.

Hands-On Activities

- Build an app with custom authentication and role-based authorization.
- Create a PWA-enabled application with push notifications.
- Implement a multi-step approval workflow using APEX Workflows.
- Create and install a custom Dynamic Action plug-in.
- Deploy an application from DEV to PROD using export/import.

Resources

- [BEENUM LEARNING – Udemy](#)
- [BEENUM LEARNING – YouTube](#)
- docs.oracle.com/en/database/oracle/apex/24.2/htmldb/

6. Phase 4: AI, Fusion Integration & Enterprise Features (Weeks 12–15)

■ PHASE 4: AI, FUSION & ENTERPRISE • Weeks 12–15

Learning Objectives

- Leverage **AI-Assisted Development (GenDev)** capabilities in APEX 24.2.
- Configure the **AI Configurations Shared Component**: system prompt, welcome message, and RAG sources.
- Use **"Generate Text with AI"** and **"Show AI Assistant"** Dynamic Actions.
- Create custom data models using **natural language** in SQL Workshop via the AI Assistant.
- Implement **Vector Search** with Oracle Database 23ai for RAG-powered applications.
- Integrate with **Oracle Fusion Cloud Applications** (ERP, HCM, SCM) via REST APIs.
- Work with **JSON Data Sources** and Oracle Database 23ai Duality Views.
- Build and manage **REST APIs** using ORDS (Oracle REST Data Services).
- Use **APEX_WEB_SERVICE** and **APEX_AI** PL/SQL APIs for programmatic integrations.
- Implement **Automations** for scheduled background tasks and data processing.
- **Performance tuning and monitoring** APEX applications for enterprise readiness.

Hands-On Activities

- Build an AI-powered app using RAG sources and the AI Assistant dynamic action.
- Create a Vector Search-enabled application with Oracle Database 23ai.
- Integrate with an Oracle Fusion Cloud REST API to fetch and display data.
- Build a JSON Data Source application using Duality Views.
- Implement an Automation for scheduled data processing and notifications.

Resources

- [BEENUM LEARNING – Udemy](#)
- [BEENUM LEARNING – YouTube](#)
- docs.oracle.com/en/database/oracle/apex/24.2/

7. Certification Path

Oracle offers official certifications to validate your expertise in Oracle APEX development.

Oracle APEX Cloud Developer Professional Certification

Exam Code: **1Z0-771**

This certification validates your ability to develop secure, scalable web and mobile applications using Oracle APEX, including AI-assisted development, Generative AI, and Oracle Database technologies.

Detail	Value
Exam Code	1Z0-771
Format	Multiple Choice
Number of Questions	55
Duration	90 Minutes
Passing Score	70%
Price	USD \$245

Exam Topic Weightings

Exam Topic	Weighting
APEX Architecture & Workspace Administration	15%
Application Development & Page Design	30%
SQL and PL/SQL Integration	25%
Interactive Reports and Forms	15%
Security and Authentication	15%

Oracle MyLearn Learning Path (22+ Hours)

Module	Topic	Duration
1	Oracle APEX Developer Professional Course	18h 34m
2	Oracle APEX: Empowering Low Code Apps with AI	3h 39m
3	Practice Exam	Self-paced
4	Online Certification Exam	Timed Exam

Also available: Oracle APEX Foundations badge (3h 42m, free) — ideal for beginners to earn a foundational credential before pursuing the professional certification.

■ docs.oracle.com/en/database/oracle/apex/24.2/education.html — Official APEX Education Hub

■ **Tip:** Focus on **Application Development & Page Design (30%)** and **SQL/PL/SQL Integration (25%)** — together they make up **55%** of the exam. Master Page Designer, Dynamic Actions, and PL/SQL processes.

8. Recommended Resources

■ Learning Courses

- **BEENUM LEARNING – Udemy** – Comprehensive Oracle APEX courses covering beginner to advanced topics.
- **BEENUM LEARNING – YouTube** – Free tutorials, walkthroughs, tips, and real-world APEX project demonstrations.

■ Official Oracle Documentation

- docs.oracle.com/en/database/oracle/apex/24.2/ – Main APEX documentation and release notes.
- docs.oracle.com/en/database/oracle/apex/24.2/htmldb/ – App Builder User's Guide (comprehensive development reference).
- docs.oracle.com/en/database/oracle/apex/24.2/aeapi/ – APEX API Reference for PL/SQL packages.
- docs.oracle.com/en/database/oracle/apex/24.2/aeutl/ – APEX Administration Guide.
- docs.oracle.com/en/database/oracle/apex/24.2/education.html – APEX Education Hub with training and certifications.

■ **Tip:** Combine BEENUM LEARNING courses with official Oracle documentation for a well-rounded learning experience. The courses provide hands-on guidance while Oracle docs serve as comprehensive reference material.

9. Suggested Practice Projects

Build these projects to solidify your knowledge and create a portfolio that showcases your APEX expertise:

1. Employee Management System

Build a full CRUD application with Interactive Reports, Forms, Charts, and a Dashboard. Implement search, filtering, and master-detail pages.

2. Project Task Tracker

Create a project management app with Workflows, approval processes, Calendar views, Gantt charts, and team collaboration features.

3. Customer Support Portal

Build a support portal with Faceted Search, email notifications, file uploads, knowledge base articles, and a customer-facing interface.

4. AI-Powered Content Assistant

Develop an app using RAG sources, AI Configurations, Generate Text with AI, and Vector Search for intelligent content generation and retrieval.

5. Expense Report Approval App

Build a multi-step approval workflow application with PWA push notifications, PDF generation for expense reports, and role-based access control.

■ **Tip:** Start with simple applications and progressively add complexity. Each project teaches different APEX capabilities — from basic CRUD to AI-powered enterprise apps. Document your work for your professional portfolio.

10. Tips for Success

1. Start with a Free Workspace

Sign up at apex.oracle.com — no installation, no credit card, no setup. You can start building apps within minutes.

2. Master SQL and PL/SQL First

SQL and PL/SQL are the backbone of every APEX application. Invest time in strengthening these skills before diving into APEX development.

3. Practice Daily

Consistency is key. Build small apps regularly to reinforce concepts and develop muscle memory with Page Designer.

4. Use Official Documentation

Refer to docs.oracle.com for the most up-to-date and comprehensive reference material on Oracle APEX.

5. Follow BEENUM LEARNING

Use BEENUM LEARNING courses on Udemy and YouTube for structured, hands-on guidance through your learning journey.

6. Learn Page Designer Inside Out

Page Designer is where you'll spend most of your development time. Master its rendering tree, properties pane, layout, and gallery.

7. Aim for the Certification

The Oracle APEX Cloud Developer Professional Certification (1Z0-771) validates your skills and adds significant value to your career.

8. Learn Incrementally

Don't try to learn everything at once. Follow the phased learning path — each phase builds on the previous one.

9. Stay Updated

Oracle APEX releases twice yearly with major new features. Follow BEENUM LEARNING on YouTube and the Oracle What's New page for the latest updates.

10. Document Your Learning

Maintain notes, code snippets, reusable templates, and lessons learned. This reference library will be invaluable throughout your career.

Best of luck on your Oracle APEX learning journey! ■

"Low-code is not about writing less code — it's about delivering more value, faster."